

Patient Churn Funnel Analysis Data Dictionary

Overview of the 3 datasets that can be joined using Patient_ID. New columns not in the original datasets are included from cleaning and exploratory data analysis processes. The reference date for the analysis is 2026-09-01.

1) patient_demographics.csv

Contains patient's information for identity, demographics, and clinical background information

Column	Type	Examples or Valid Range/Values	Description	Notes
Patient_ID	Text	P0002, P0159	An unique identification number for a patient; the primary key for joining the 3 datasets	830 rows, 805 unique values, and has duplicate values that need to be filtered out
First_Name	Text	Charlotte, reBecca	The patient's first name	Casing inconsistency that needs standardizing
Last_Name	Text	Lopez, lee	The patient's last name	Casing inconsistency that needs standardizing
Age	Number	0 - 90	The patient's recorded age in years	Invalid values like negatives and 999 that need to be filtered out, values that are blank or 0 are treated as Unknown rather than being dropped
Age_Groups	Text	Adolescent, Young Adult, Middle Age,	The age group that a patient's age fall into, based on Age	Conditions for each age group are: Age>0 falls

		Senior, Unknown	column	into "Adolescent", Age>18 falls into "Young Adult", Age>40 falls into "Middle Age", Age>60 falls into "Senior", and if Age is blank or) fall into "Unknown"
Gender	Text	Female, Male	The patient's recorded gender	Casing inconsistency that needs standardizing
State	Text	CA, FL, IL, NY, TX, ...	The patient's US state of residence as a 2 letter abbreviation	20 different states
Primary_Diagnosis	Text	Arthritis, Asthma, COPD, Diabetes, Hypertension, ...	The patient's main diagnosis	21 different diagnosis, misspellings and abbreviations that need to be corrected to collapse into other valid values
Department_Specialty	Text	Cardiology, Endocrinolog y, General Practice, Pulmonology, ...	The department that the patient is under	10 different departments, misspellings and abbreviations that need to be corrected to collapse into other valid values
Insurance_Provider	Text	Aetna, BlueCross/BC BS, Cigna, Medicare,	The patient's insurance provider	5 different insurance carriers, where Medicare is

		UnitedHealth		assigned to patients with Age > 60
Out_of_Pocket_YTD	Number	0 - 4999	The patient's out of pocket spending in USD, year to date	Higher cost departments tend to have a higher value

2) patient_appointment_metrics.csv

Holds digital engagement, visit history, and churn status of patients

Column	Type	Valid Range/Values	Description	Notes
Patient_ID	Text	P#####, where # is a number	An unique identification number for a patient; can join with the other 2 datasets	830 rows, 805 unique values, and has duplicate values that need to be filtered out
Referral_Sources	Text	Online Search, Zocdoc, Physician Referral, Friend Referral, Social Media	How the patient was referred or aquired	5 different categories
Website_Visits	Number	1 -13	The number of times a patient visits the website	No nulls
Page_Views	Number	2 - 74	The number of times a page on the website is viewed by a patient	Scales with Website_Visits
Booking_Time_Mins	Number	1.0 - 30.6	The amount of time in minutes a patient spends on booking the appointment	No nulls

Avg_Wait_Time_Mins	Number	5 - 60	The amount of time in minutes a patient spends in the waiting room before their appointment	No nulls
Avg_Appointment_Time_Mins	Number	10 - 47	The amount of time in minutes of a patient's appointment	No nulls
Total_Visits	Number	1 - 20	The total number of visits a patient has had	35 missing values
Missed_Appointments	Number	0 - 5	The total number of missed appointments a patient has had	No nulls
Last_Visit_Date	Date	2025-06-30	The date of the last visit the patient has had	Mixed formats that need standardizing, 43 missing values, used as basis for Churn_Status_Updated
Satisfaction_Score	Number	1 - 10 (low to high)	A patient's satisfaction score of their overall experience	Correlated to churn, where lower scores go with churned patients and higher scores align with active patients
Churn_Status_Updated	Text	Yes, No	Indicator of whether or not a patient has or has not been churned	Values are updated using Last_Visit_Date and the reference date, where no visit in

				2 or more years or no date is recorded means that the patient has been churned
Churn_Status_Boolean	Boolean	1, 0	Boolean indicator of whether or not a patient has or has not been churned	Based on Churn_Status_Updated, where 1 means a churned patient and 0 means an active patient

3) patient_contact.csv

Includes contact information and outreach/response history

Column	Type	Valid Range/Values	Description	Notes
Patient_ID	Text	P#####, where # is a number	An unique identification number for a patient; can join with the other 2 datasets	830 rows, 805 unique values, and has duplicate values that need to be filtered out
phone_number	Text	737.702.6699, 268-632-7015, (878) 573-3758	The patient's phone number, in ###-###-#### format	Inconsistent separators that need standardizing, 50 missing values
email	Text	CHARLOTTE.GARCIA@YAHOO.COM Rebecca.Lopez@outlook.com	The patient's email address	Casing inconsistency that needs standardizing, 33 missing values
last_patient_outreach_date	Date	2025-08-18	The date of the last time a patient has been contacted, in	Mixed formats that need standardizing, 35 missing

			yyyy - mm - dd format	values
last_patient_response_date	Date	2025-08-26	The date of the last time a patient has reached out, date, in yyyy - mm - dd format	Mixed formats that need standardizing, 209 missing values